Harkunwar Singh

Hamilton, ON • 365-889-3885 • singhharkunwar87@gmail.com • LinkedIn • GitHub • Portfolio

Summary

- Honours Computer Science Co-op graduate from McMaster University (April 2026) with a 4.0 GPA in final year, available immediately for full-time Software Engineering roles.
- Developer with experience building production-style backend, mobile, data-processing, and ML systems using Python, SQL, Java, TypeScript, FastAPI, PostgreSQL, and cloud platforms, backed by a year-long government co-op at the Ministry of Transportation Ontario doing QA testing, defect lifecycle management, and Python process automation on a live enterprise system.
- Hands-on experience with relational databases, REST APIs, automated testing, Git, Docker, CI/CD workflows, Jira, Confluence, and applied machine learning.

Education

McMaster University Hamilton, ON

Honours B.Sc., Computer Science (Co-op) *Sept. 2021 - Apr. 2026*

- **Award:** McMaster Faculty of Engineering Award of Excellence Scholarship for Academic Excellence ($3,000).
- **GPA:** Achieved a 4.0 out of 4.0 in the final year of the Honours Computer Science program.
- **Relevant Coursework:** Data Structures and Algorithms, Software Development Life Cycle, Operating Systems, Information Security, Data Analytics, Computer Architecture, Web Programming, Statistics, Discrete Mathematics.

Skills

Languages: Python, SQL, Java, C/C++, JavaScript, TypeScript, Assembly, HTML, CSS
Frameworks and Libraries: FastAPI, SQLAlchemy, Pandas, Playwright, NumPy, Scikit-learn, PyTorch, React Native, Expo, Node.js, Streamlit, LangChain
Databases and Cloud: PostgreSQL, MySQL, MongoDB, ChromaDB, Supabase, AWS S3, Microsoft Azure
Tools and Platforms: Git, Docker, Linux, Azure DevOps, Jira, Confluence, Jest, Pytest, Wireshark, Figma
AI Tools: GitHub Copilot, Claude Code, ChatGPT Codex, Cursor, Claude Cowork, Ollama
Technical Areas: REST APIs, Relational Databases, Data Processing, ETL Concepts, QA Testing, CI/CD, Object-Oriented Design, Machine Learning, NLP, Data Structures and Algorithms

Professional Experience

Engineering Associate Co-op North York, ON

Ministry of Transportation Ontario *Aug. 2024 - Aug. 2025*

- Performed end-to-end QA testing and functional validation for a live government Laboratory Information Management System (LIMS) across 5+ release cycles, identifying defects and edge cases before production deployment.
- Managed the full bug lifecycle in Azure DevOps, creating, tracking, and validating tickets through resolution, and coordinating defect fixes with an external vendor development team.
- Developed a Python data-processing workflow using pdfplumber and pandas to extract, clean, validate, and convert laboratory-result tables from PDFs into standardized CSV files for LIMS ingestion, eliminating hours of manual data entry across 3+ government lab departments.
- Coordinated QA and testing efforts across 3+ government departments, aligning functional requirements and communicating defect status to the vendor team throughout multiple release cycles.
- Authored bug ticket templates, testing guides, and reporting procedures used for team standardization and onboarding across the QA process.

Projects

NexDerm `[GitHub]` | *React Native, TypeScript, FastAPI, AWS S3, PostgreSQL, PyTorch, SQLAlchemy* Sept. 2025 - Apr. 2026

- Co-developed a cross-platform AI skin lesion screening application for iOS and Android using React Native, Expo, and TypeScript, supporting image capture, real-time predictions, doctor review flows, and PDF report generation.
- Engineered a FastAPI backend deployed on AWS with REST endpoints for authentication, image upload, ML inference, report generation, and doctor review, using SQLAlchemy for database ORM and Supabase PostgreSQL for relational data persistence.
- Integrated AWS S3 for scalable image storage with IAM-secured bucket policies and Resend for transactional email delivery, supporting a complete end-to-end data and communication flow.
- Trained DenseNet and ResNet CNN models using PyTorch on a 16-class Kaggle skin lesion dataset, combining predictions through an ensemble approach to achieve approximately 70% overall accuracy, with top classes reaching 85-90%.
- Implemented a pytest-based backend test suite with coverage reporting and Jest-based frontend tests, with external services stubbed for deterministic test execution.
- Used Jira and Confluence to assign development work, document requirements, coordinate Git branches, and track project progress across the team.

Quill `[GitHub]` | *Python, LangChain, Ollama, LLaMA 3.2, ChromaDB, Streamlit* Mar. 2026 - May 2026

- Designed and implemented a fully offline RAG application that lets users chat with any PDF using local LLMs, with no API keys or external data transmission, ensuring complete data privacy.
- Created a document ingestion pipeline using RecursiveCharacterTextSplitter with 1000-token chunks and 200-token overlap, embedded with nomic-embed-text via Ollama, and stored in ChromaDB for persistent local retrieval.
- Applied MMR retrieval to fetch the top-5 most relevant and diverse chunks from 20 candidates, reducing redundancy and improving answer quality across multi-section documents.
- Integrated LLaMA 3.2 via Ollama to generate grounded answers with page-level citations, and evaluated pipeline quality using RAGAS metrics including faithfulness and answer relevancy.

AI Resume-Job Matching System `[GitHub]` | *Python, TF-IDF, Scikit-learn, NLP, Pandas* Jan. 2026 - Present

- Created an explainable NLP-based resume matching system that compares candidate resumes against software engineering job descriptions using TF-IDF vectorization and cosine similarity.
- Designed a preprocessing pipeline with tokenization, stop-word removal, n-grams, and skill extraction using 100+ real-world job postings to improve keyword coverage and reduce noisy matches.
- Developed a scoring module that returns match percentages, matched skills, missing keywords, and prioritized resume improvement suggestions for non-technical users.